

Smarter SLO: Anomaly Detection for Infrastructure with Elastic



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Overview

- The Silent Problems with static thresholds
- SLI, SLO and Error Budget ?
- What is anomaly detection?
- Our ML approach
- Architecture
- Live demo
- Expected outcome
- Q & A



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THE PROBLEM

Static thresholds can't protect modern SLOs

1. Fixed lines (“CPU > 90%”) are too slow
2. By the time one fires, error budget is often already spent
3. Every host and service group has a different “normal”
4. Daily and weekly patterns trigger false alarms
5. Hundreds of hosts means hundreds of thresholds to tune
6. Slow drift goes unnoticed until it becomes an incident

What we need instead

A way to learn what “normal” looks like for every host and service group — and flag the moment something drifts, with enough lead time to act before the SLO breaches.

Example: “CPU > 90%” fires every morning at market open (normal), yet misses a host that quietly climbs from 30% to 70% over a week.

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FOUNDATIONS

SLI, SLO and Error Budget

SLI

Service Level Indicator

The measured ratio: good events ÷ total events. e.g. % of memory readings ≤ 80% used.

SLO

Service Level Objective

The target you commit to for that SLI. e.g. “99% of readings should be good.”

Error Budget

1 − objective

The allowed amount of “bad.” A 99% objective = 1% budget before you're in violation.

Burn Rate

How fast you're spending it

1.0 = sustainable pace. 200x = you'll exhaust a month's budget in hours.

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How does anomaly detection solve this ?





ANOMALY DETECTION

What is anomaly detection?

It automatically spots behavior that deviates from what is normal for your data — without anyone having to write the rule.

1 Learn

Models each metric's normal behavior over time, including daily and weekly patterns.

2 Compare

Checks every new data point against the learned model in near real time.

3 Score

Gives each deviation an anomaly score (0–100), so you focus on the most unusual first.

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TYPE OF LEARNING

Unsupervised learning: no labels, no rules

Supervised

Needs labeled examples of past incidents, and can only recognize problems it has already seen.

Not what we use here

Unsupervised — Elastic anomaly detection

- Learns from raw, unlabeled metrics — no incident history required
- Builds a statistical model of “normal” per entity, including seasonality and trends
- Flags what is unusual, including problems never seen before
- Keeps adapting as new data arrives

THE ANSWER

How anomaly detection fixes the threshold problem

STATIC THRESHOLDS	WITH ANOMALY DETECTION
Too noisy or too slow	Alerts on how unusual a value is for that entity, not on one fixed number
Error budget already spent when it fires	Flags drift early — the gap before the SLO breaches is your lead time
Every host has a different “normal”	Learns a separate baseline per host and per service group
Slow, subtle drift goes unseen	Catches gradual change, like a memory leak, long before it crosses a line

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Why not just a basic anomaly job ?



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OUR APPROACH · STEP 1

Starting point: the basic anomaly detection job

What a basic job does

- Models one metric as a single time series
- Learns normal over time and flags spikes or dips
- Quick to set up — a good first step

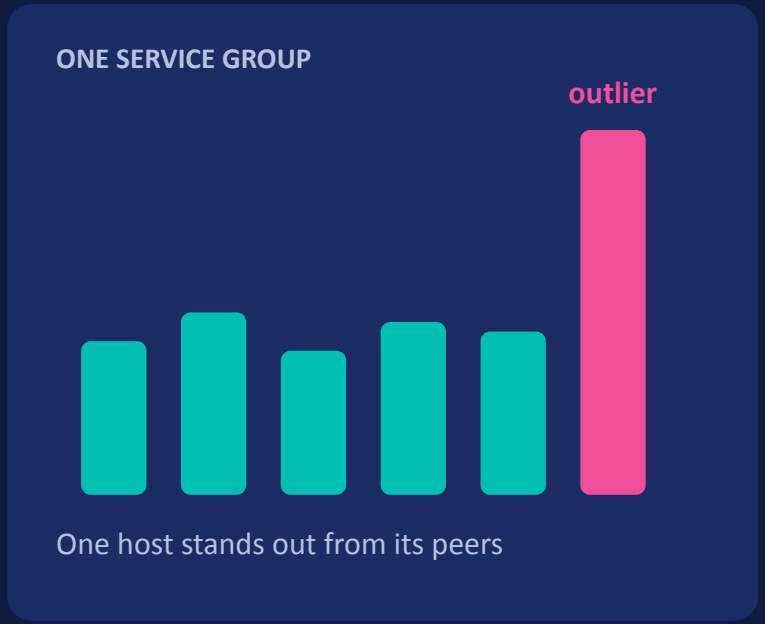
Where it falls short for a fleet

- The whole fleet blends into one line — one bad host gets averaged out
- Ignores that service groups behave differently
- Can't tell you which host is the problem

So we didn't jump straight to the final design — each step fixes a gap in the one before.

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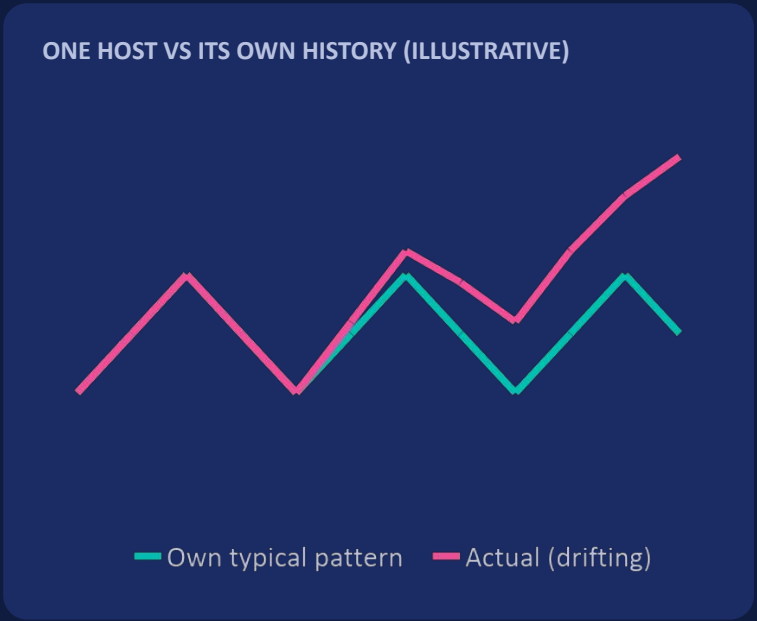
Why we added population analysis



- Compares each host against its peers in the same service group
- Flags the one host that stands out — no long per-host history needed
- New hosts are covered from day one

Limit: a host is judged only against its peers, so a server with a unique pattern can be misjudged.

Why trading hosts get their own baseline



- Compares each server against its own history
- Used for high-criticality trading hosts, where each server's own normal matters more than peer comparison
- Catches slow drift that peer comparison alone would miss

Trade-off: it needs history to learn from, so it complements the fleet track rather than replacing it.

by + partition vs over + partition

```
{
  "function": "high_mean",
  "field_name": "system.cpu.total.norm.pct",
  "by_field_name": "host.name",
  "partition_field_name": "service.common_name"
}
```

Per-host baseline

by + partition

Each host vs its own history, with a separate model per service group.

```
{
  "function": "high_mean",
  "field_name": "system.cpu.total.norm.pct",
  "over_field_name": "host.name",
  "partition_field_name": "service.common_name"
}
```

Fleet population

over + partition

Each host vs its peers in the same service group, with a separate model per group.

OUR APPROACH

Putting it together: two tracks in one job per metric

	Basic job	Fleet population	Per-host baseline
Compares against	The overall trend of one metric	Peers in the same service group	The host's own history
Best at	A quick first look	Spotting the outlier host	Drift on critical trading hosts
Gap	Hides which host is at fault	Unique patterns, group-wide shifts	Needs history to learn from

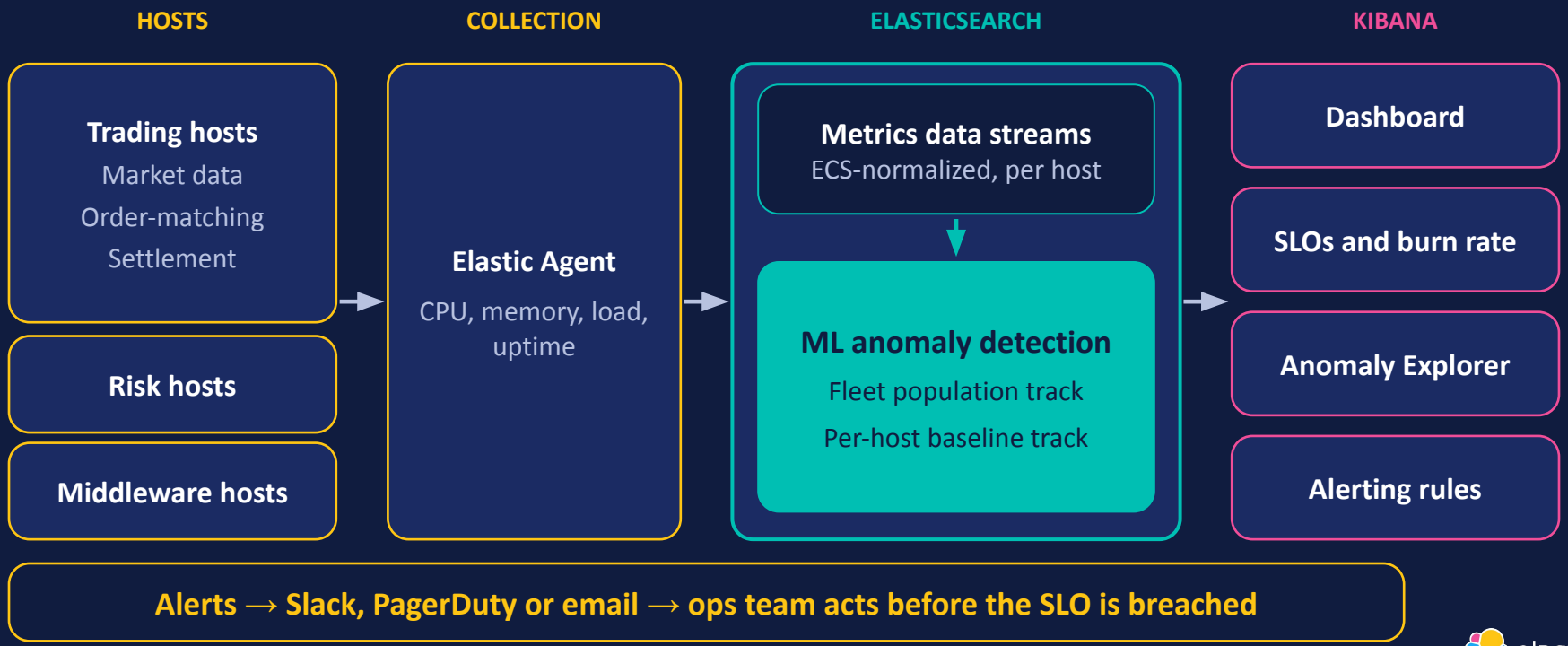
Every metric runs both tracks together

CPU · Memory · System load · Host availability

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ARCHITECTURE

From host metrics to early warning



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ANOMALY EXPLORER · SINGLE METRIC VIEWER

Reading the anomalies in Kibana

Anomaly Explorer

Start here: where and when



Swim lanes: warmer color = higher score

- Swim lanes by job and by host.name
- Colored by anomaly score, 0–100
- Click a cell to drill into the anomalies

Single Metric Viewer

Then zoom in: one metric over time



Illustrative: actual leaves the expected range

- Actual value vs expected range
- Anomalies marked on the timeline
- Pick the host and time range to inspect

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Live Demo

Pause the slides here and switch to Kibana

1. **Create an anomaly detection job live**
2. **See the anomalies it flags**
3. **Walk through the dashboard**

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EXPECTED OUTCOME

From reactive to proactive SLO management



Know where it started

An SLO tells you a breach happened; ML shows which host and metric drifted first.



Two lenses, full coverage

Fleet detectors catch the outlier host; per-host baselines catch drift peers would miss.



Time to act

The gap between “anomaly flagged” and “SLO violated” is the window ops needs.



One framework

The same pattern for CPU, memory, system load and availability across the fleet.

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Thank you

Questions?

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